



Infection Prevention and Management Manual

Safe Injection Principles Policy

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation-Wide	Clinical staff that handle and administer injectable medications

Introduction

Transmission of bacteria and viruses may occur via unsafe injection practices resulting in serious complications. This may arise from extrinsic contamination of products, vials, re-use of syringes and medications.¹

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☐	Staff Safety	☐
Other:	☐		

Policy Principles

This policy outlines the expected clinical practice standards for the preparation and administration of injections to minimise the likelihood of transmission of infectious diseases either between patients or between a patient and healthcare worker.

1.0 General injection practices

The principals of standard precautions, including aseptic technique, hand hygiene and safe management and disposal of sharps always apply when preparing and administering injections. This includes:

- The use of sterile, single-use needles, cannulas, syringes and giving sets. Once used, these items are considered contaminated and must be discarded.²
- Using single dose vials, ampoules or pre-filled syringes for one patient only.^{1,2}
- Medications or solutions that are in contact with sterile tissues must remain sterile.²
- Ensuring medication from the same syringe are not administered to more than one patient, even if the needle is changed or the injection is administered via an intervening length of IV tubing.^{1,2}
- Except for insulin, multi-dose vials must not be used.² Multi-dose insulin vials must be dedicated to a single patient, labelled with patient details and date/time of opening and managed as per SCGOPHCG [Nursing Practice Guideline for Diabetes Patient Management](#) and [High Risk Medication](#). Alternative indications for multi-use vials must have a documented procedure for preparation and management with

endorsement from the SCGOPHCG Medicines and Therapeutics Committee. This is regardless of medications identified by manufacturer labelling as a Multi-dose.²

- Use of available safety-engineered medical devices.^{1, 2}
- Discarding medication or injection materials if sterility is compromised.²
- Discarding sharps directly into a hospital designated sharps disposal container, sharp first, as close to the point of use and as soon as possible.
- Disposing used injection equipment into the appropriate waste stream (Refer to SCGH and OPH Hospital Policy for Waste available on [HealthPoint](#)).

In the event of a needle stick injury at SCGH or OPH, refer to [SCGOPHCG Infection Prevention and Management Manual - Prevention and Management of Sharps Injuries and Body Substance Exposure](#).

1.1 Preparation of medication for injection

- Draw up medications and prepare infusions in a designated clean medication preparation area that is not adjacent to potential sources of contamination, including sinks or other water sources.² Promote aseptic technique by ensuring supplies, e.g., alcohol-based hand rub, sterile needles and syringes, disinfectant wipes and impregnated clean and disinfectant wipes, are readily available.
- Ensuring adequate lighting and minimal distraction.¹
- Ensure preparation surfaces are clean and free of visible contaminants such as blood, body fluids or other equipment.¹ This applies to general work surfaces, stainless steel procedure trolleys, disposable single-use kidney dishes and re-usable procedure trays.
- Prepare injections and infusions as close as possible to the time of administration. This reduces the risk of microbial contamination or bacterial proliferation and helps to maintain the physical and chemical stability of the medication.²
- Do not leave a needle inserted into a medication vial septum for multiple uses or later access.
- Consider the availability and use of manufactured pre-filled syringes / infusions.
- The vial cap acts only as a shield or dust cover for the vial septum, protecting it from dust and other physical contamination. The vial septum is not sterile.³ Use a sterile single-use 70% alcohol-impregnated wipe to scrub the vial septum. Scrub the vial septum for 30 seconds using different areas of the wipe. Allow to dry. This practice may vary for vaccines – refer to [The Australian Immunisation Handbook](#).⁵
- All medication vials must be accessed with a new sterile needle and syringe and the vial septum scrubbed as above even when obtaining additional doses for the same patient.¹
- If a single-dose vial has been opened or accessed, the vial is to be discarded at the end of the procedure for which it is being used, or according to the duration the manufacturer specifies, whichever comes first. It should not be stored for future use, even on the same patient.^{1,2}
- Never combine leftover medications from different vials for later use.^{1,2}
- Do not use bags or bottles of intravenous solution as a common source of supply for multiple patients.²
- Fluid infusion and administration sets must be used for one patient only and discarded appropriately after use.²
- Administration sets are single use devices and if disconnected from the intravenous (IV) device for any reason e.g., intermittent medication dosing, the set is to be discarded and a new administration set connected using aseptic technique.

2.0 Regional Anaesthesia and Vascular Access Procedures

There is a risk of accidental injection of chlorhexidine when used as a skin antiseptic for regional anaesthesia and vascular access procedures. Maintaining patient safety and preventing this risk requires the following:

- Unless contraindicated and where possible, use 2% chlorhexidine and 70% alcohol as a tinted solution when antiseptics is required for skin preparation before administering regional anaesthesia or prior to a vascular access procedure.⁴
- If topical chlorhexidine solution is required, it must be dark tinted to provide a visual cue that the solution is non-injectable.⁶
- Skin preparation to be completed, and topical antiseptic discarded before exposing sterile equipment and injectable fluids on the sterile procedure area. The skin antiseptic solution should remain outside the sterile area during the procedure. Very small quantities of neuraxial chlorhexidine have been implicated in cases of severe neurotoxicity.⁶
- Non-injectable fluids, including chlorhexidine, should never be decanted into an open container on a sterile procedure area.⁶
- Injectable medicines should not be decanted into a gallipot or other open container before preparation and injection; if this is unavoidable, any medicine that is decanted should be immediately drawn up and labelled.⁶
- When centroneuraxial procedures are being performed, such as a lumbar puncture or insertion of catheter into the epidural space, surgical aseptic technique and maximal barrier precautions are required. Bacterial meningitis has occurred following injection of material or insertion of a catheter into epidural or subarachnoid spaces.⁴

Compliance, monitoring, and evaluation

Compliance against this policy will be evaluated via routine clinical incident management review processes.

Related Documents

- SCGOPHCG Clinical Practice Guideline. Peripheral Intravenous Catheters (PIVC) Insertion, Maintenance and Removal
- SCGOPHCG Infection Prevention and Management Manual:
 - Standard and Transmission-Based Precautions
 - Prevention and Management of Sharps Injuries and Body Substance Exposure.
 - Aseptic Technique
- SCGOPHCG Medicines Management Policy and Procedure
- SCGOPHCG High Risk Medication Policy
- SCGOPHCG Nursing Practice Guideline. Diabetes Patient Management
- SCGOPHCG Hospital Wide Policy. Waste, Disposal and Environmental Management #HP079

References

1. Dolan SA, Meehan KA, Felizardo G, Barnes S, Kraska S, Patrick M, et al. APIC position paper: Safe injection, infusion, and medication vial practices in health care. Am J Infect Control [Internet]. 2016 [cited 2024 November 12]; 44(7):750-7: Available [https://www.apic.org/Resource /TinyMceFileManager/Position_Statements/2016APICSIPP ositionPaper.pdf](https://www.apic.org/Resource/TinyMceFileManager/Position_Statements/2016APICSIPP ositionPaper.pdf)
2. CDC. Preventing Unsafe Injection Practices. 2024. [Cited 2025 April 2] Available <https://www.cdc.gov/injection-safety/hcp/clinical-safety/index.html>

3. National Health and Medical Research Council. Australian guidelines for the prevention and control of infection in healthcare, Canberra: National Health and Medical Research Council. [Internet]. 2019. [Cited 2024 November 12] Available <https://www.nhmrc.gov.au/about-us/publications/australian-guidelines-prevention-and-control-infection-healthcare-2019>
4. Australian and New Zealand College of Anaesthetists. Guidelines on Infection Control in Anaesthesia. PS 28 [Internet]. 2015. [Cited 2020 May 26] Available [PG28\(A\) Guideline on infection control in anaesthesia 2015.PDF](#)
5. Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook, Australian Government Department of Health, Canberra, [Internet]. 2015. [Cited 2024 November 12] Available <https://immunisationhandbook.health.gov.au/>
6. Australian and New Zealand College of Anaesthetists. Australian Commission on Safety and Quality in Health Care. Joint safety statement. Topical application of chlorhexidine and the risks of accidental injection in regional anaesthesia and vascular access procedures. [Internet]. 2018. [Cited 2024 November 12] Available <https://www.safetyandquality.gov.au/our-work/medication-safety/position-and-safety-statements#:~:text=A joint safety statement was developed in response,national anaesthetic and user-applied labelling guidelines and standards.>

Authorisation & Version Control

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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